Ship Shape

Water pushes up on anything in it. A barge floats as long as it can push aside water that weighs as much as the barge and its cargo, so shape matters as much as material.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 23	Foil Boat Challenge
Lesson 2	Fri Dec 4	First Kit Barge
Lesson 3	Fri Jan 22	Design Duel and First Estimate
Lesson 4	Fri Feb 19	Mock Tournament
Lab I1	Before Lesson 2	The Push of the Water (home or an open Friday)
Lab I2	Before Lesson 3	Archimedes, Density, and Folding (home or an open Friday)
Lab I3	Before Lesson 4	Estimate and Score Practice (home or an open Friday)

At the tournament

Pairs get 15 minutes to build a barge from a kit (foil, plastic sheet, 50 cm of tape, 5 straws) that fits in a 15 x 15 x 2.5 cm box, estimate its cargo to 0.01 g, load hex nuts until it sinks, and take a written test on buoyancy (bring goggles, scissors, a ruler, pencils; calculator optional).

Keep every handout for this event behind this divider, in order. Bring this binder to every Build & Test Fridays meeting.

Ship Shape: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can explain the event: a 15-minute barge build, an estimate, a cargo test, and a written test.	L1 L4	☐	☐
I can list the kit materials and the barge rules: one open cargo area, fits in a 15 x 15 x 2.5 cm box, only kit materials.	L1 L2 L4	☐	☐
I can wear Category C indirect-vent goggles when testing and keep water spills safe.	L1 L4	☐	☐
I can define buoyancy (buoyant force) and describe the upward push of water.	L4 I1 I2	☐	☐
I can define displacement and measure it in mL with a measuring cup.	L4 I1	☐	☐
I can state Archimedes' principle: the upward push equals the weight of the water pushed aside.	L2 L4 I2	☐	☐
I can explain flotation: an object floats when the water it pushes aside weighs as much as it does before it goes under.	L4 I1 I2	☐	☐
I can explain why a steel ship floats but a steel nut sinks, using shape and density.	L1 L4 I2	☐	☐
I can use 1 cubic cm of water = 1 mL = 1 gram to predict how much cargo a barge can hold.	L2 L3 I2 I3	☐	☐
I can build a watertight barge from the kit in 15 minutes or less.	L2 L3 L4	☐	☐
I can check my barge with the 15 x 15 x 2.5 cm measuring box.	L2 L3 L4	☐	☐
I can load cargo one piece at a time and spread it evenly so the barge stays level.	L3	☐	☐
I can make a cargo estimate to 0.01 g using my barge's size, my practice results, and the mass of one nut.	L3 L4 I3	☐	☐
I can explain the score: test points plus grams held plus a 5-point bonus for an exact estimate.	L3 L4 I3	☐	☐

Foil Boat Challenge

TODAY I CAN

- I can explain the event: a 15-minute barge build, an estimate, a cargo test, and a written test.
- I can list the kit materials and the barge rules: one open cargo area, fits in a 15 x 15 x 2.5 cm box, only kit materials.
- I can wear Category C indirect-vent goggles when testing and keep water spills safe.
- I can explain why a steel ship floats but a steel nut sinks, using shape and density.

Welcome to Ship Shape! You build a barge (a flat-bottomed boat that carries cargo) and see how much it can hold before it sinks.

Today's steps: run it yourselves (22 minutes)

You need: 4 foil squares (12 x 12 in), cups of hex nuts (50 per cup), 4 pairs of Category C goggles, strainer and towels, 2 paper sheets with a 15 x 15 cm square drawn on them. Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read 'The event in a nutshell' out loud. Each partner says one rule back from memory.
2. Both 5 min. Set the timer for 5 minutes. Shape one foil square into a boat that fits on your 15 x 15 cm paper square. Fold, do not crumple. No tape today.
3. Both 5 min. Goggles on. Set the timer for 5 minutes. Get an adult in the room to say 'Clear!' before testing. Set your boat in the tub and add nuts one at a time, gently. Count until water pours in, and fill row 1 of 'Our foil boats'.
4. Both 6 min. Set the timer for 6 minutes. Make boat 2 with one change (taller sides, flatter bottom, or stronger corners). Test it the same way (goggles on, adult says 'Clear!') and fill row 2.
5. Both 3 min. Set the timer for 3 minutes. Answer 'Think about it', then check with the answer card. Strain the nuts and wipe up spills.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Compare your 'Think about it' answers to the card and add one idea you missed in a different color.

The event in a nutshell

Kit	1 foil sheet (12 x 12 in), 1 plastic sheet (10.75 x 10 in), 50 cm of masking tape, 5 straws.
Build time	15 minutes at the tournament.
Size rule	The barge must fit inside a box 15 cm x 15 cm x 2.5 cm tall.
Cargo	Metal hex nuts, added one at a time until the barge sinks (or holds 500 grams).
Test	A written test on floating and sinking.
Safety	Wear Category C indirect-vent goggles while testing.

Our foil boats

Boat	Shape (draw or describe)	Nuts held	What made it sink?
1			
2			

Think about it

1. Which boat held more? What was different about its shape?

2. A nut sinks, but a foil boat full of nuts can float. Why do you think that is?

At-home practice

Answer these at home. Check them with the answer card at the next Huddle.

1. How many minutes do you get to build your barge at the tournament?

2. What is in the tournament kit?

- ◯ A 1 foil sheet, 1 plastic sheet, 50 cm of masking tape, 5 straws
- ◯ B Foil, glue, clay, and paper cups
- ◯ C Cardboard, craft sticks, and string

3. Which goggles must you wear when you test?

- ◯ A Regular safety glasses
- ◯ B Category C indirect-vent goggles
- ◯ C Your own eyeglasses

4. A foil sheet squeezed into a tight ball sinks. The same sheet shaped into a boat floats. What changed?

5. What is the cargo at the tournament, and how is it added?

At home before next time: Answer the At-home practice questions on this handout. Do Independent Lab I1 (The Push of the Water) before Lesson 2. It starts with a Sink or Float test like the guesses you made today. Engineer's Logbook: build a Penny Barge (p. 13), then beat it with a new shape, for the Penny Loader badge.

First Kit Barge

TODAY I CAN

- I can list the kit materials and the barge rules: one open cargo area, fits in a 15 x 15 x 2.5 cm box, only kit materials.
- I can state Archimedes' principle: the upward push equals the weight of the water pushed aside.
- I can use 1 cubic cm of water = 1 mL = 1 gram to predict how much cargo a barge can hold.
- I can build a watertight barge from the kit in 15 minutes or less.
- I can check my barge with the 15 x 15 x 2.5 cm measuring box.

Today you build your first barge from the real kit. Measure it carefully: you will use Archimedes' rule to predict how much it can hold.

Today's steps: run it yourselves (22 minutes)

You need: 2 full practice kits, 15 x 15 x 2.5 cm measuring box, about 200 hex nuts, kitchen scale, 4 pairs of Category C goggles, strainer and towels, 1 cm cube, calculator. Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read 'The big rules' out loud. Hold up the 1 cm cube: that much water is 1 mL and weighs 1 gram.
2. Both 12 min. Set the timer for 12 minutes. Build a barge from the kit using only kit materials, scissors, and a ruler. Fold the corners; do not cut them. The Event Judge calls 10, 5, and 1 minute left.
3. Both 1 min. Put your barge in the measuring box. It must fit all the way inside.
4. Both 5 min. Goggles on. Set the timer for 5 minutes. Get an adult in the room to say 'Clear!' before testing. Load nuts one at a time until the rim goes under. Strain the nuts, weigh them on the kitchen scale, and fill 'Nuts held' and 'Actual cargo held'.
5. Both 1 min. Measure your barge's inside length, width, and wall height and write them in 'Capacity math'. Answer 'Think about it' at home.

Check your work: Ask your Event Captain for the Lesson 2 answer card. Check your box fit and your Do Now volume. You finish the capacity math in Independent Lab I2.

The big rules

Archimedes' principle	The upward push on an object equals the weight of the water it pushes aside.
The gram rule	1 cubic cm of water = 1 mL = 1 gram.
Volume	Length x width x height. Measure in cm to get cubic cm.
Capacity	Volume of the hull below the rim (in g) minus the mass of the barge = most cargo it can hold.

Capacity math

Step	Our barge
Inside length (cm)	
Inside width (cm)	
Wall height (cm)	
Barge mass (g), if there is time to weigh it	
Nuts held	
Actual cargo held (g)	

Think about it

1. Where did water come in first? What will you change next time?

Sketch our barge and label where the tape went

At-home practice

Answer these at home. Check them with the answer card at the next Huddle.

1. Fill in: 1 cubic cm of water = ___ mL = ___ g.

2. A hull is 8 cm x 8 cm x 2 cm. What is its volume? How many grams of water can it push aside?

3. Finish Archimedes' principle: the upward push on an object equals ...

4. The barge must fit inside a box that is:

◯ A 15 cm x 15 cm x 5 cm ◯ B 25 cm x 25 cm x 2.5 cm ◯ C 15 cm x 15 cm x 2.5 cm

5. Why is it better to fold the foil at the corners instead of cutting it?

At home before next time: Answer the At-home practice questions on this handout. Do Independent Lab I2 (Archimedes, Density, and Folding) before Lesson 3. Keep this page: the lab uses your barge's measurements and cargo held. Engineer's Logbook: do the Capacity Predictor (p. 14), predicting with 1 cubic cm = 1 g before you test, for the Archimedes Ace badge.

Design Duel and First Estimate

TODAY I CAN

- I can use 1 cubic cm of water = 1 mL = 1 gram to predict how much cargo a barge can hold.
- I can build a watertight barge from the kit in 15 minutes or less.
- I can check my barge with the 15 x 15 x 2.5 cm measuring box.
- I can load cargo one piece at a time and spread it evenly so the barge stays level.
- I can make a cargo estimate to 0.01 g using my barge's size, my practice results, and the mass of one nut.
- I can explain the score: test points plus grams held plus a 5-point bonus for an exact estimate.

Engineers test two ideas the same way to find the better one. Today your team tests two barge designs and writes an estimate before each test, just like the tournament.

Today's steps: run it yourselves (27 minutes)

You need: 2 full practice kits, measuring box, about 200 hex nuts, 0.01 g digital scale, kitchen scale, 4 pairs of Category C goggles, strainer and towels, calculator. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Weigh 10 dry nuts on the 0.01 g scale and divide by 10. Fill the first two rows of 'Estimate worksheet'. Pick Design A or Design B; the two pairs pick different designs.
2. Both 12 min. Set the timer for 12 minutes. Build your design. The Event Judge calls 10, 5, and 1 minute left.
3. Both 1 min. Box check: put your barge in the measuring box. It must fit all the way inside.
4. Both 3 min. Set the timer for 3 minutes. Weigh your barge and fill 'Estimate worksheet' down to 'Estimate'. Use the real-world percent from Independent Lab I2. Write the estimate before testing.
5. Both 8 min. Goggles on. Set the timer for 8 minutes. Get an adult in the room to say 'Clear!' before testing. Load both barges the same way: middle first, then corners and edges evenly. Count, strain, weigh, and fill 'Design duel' and the last rows of the worksheet. Answer 'Think about it'.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Compare each worksheet step to the example math on the card. Your numbers will be different, but every step should match. Check 'Think about it' and fix any you missed in a different color.

Design duel

Design	Materials used	Build time (min)	Nuts held	Mass held (g)
A				
B				

Estimate worksheet

Step	Our numbers
Mass of 10 nuts (g)	
Mass of 1 nut = mass of 10 / 10 (g)	
Inside L x W x wall height (cubic cm)	
Minus barge mass (g)	
Real-world percent from Lab I2 (%)	
Predicted cargo (g)	
Predicted nuts = predicted cargo / mass of 1 nut	
Estimate = nuts x mass of 1 nut (to 0.01 g)	
Actual mass held (g)	
Difference (g)	

Think about it

1. Which design held more? What do you think made the difference?

2. Why should you spread nuts evenly instead of piling them in one spot?

3. How close was your estimate? What would you change in your math next time?

At-home practice

Answer these at home. Check them with the answer card at the next Huddle.

1. 10 nuts have a mass of 30.80 g. What is the mass of 1 nut?

2. You think your barge will hold 50 of those nuts. What is your estimate to 0.01 g?

3. A cargo area is 12 cm x 10 cm x 2 cm and the barge has a mass of 10 g. What is the most cargo it could hold?

4. Why should the nuts go in the middle first, then spread out evenly?

5. Which estimate is written to the 0.01 g, like the score sheet asks?

○ A 187 g ○ B 187.46 g ○ C 187.5 g

At home before next time: Answer the At-home practice questions on this handout. Do Independent Lab I3 (Estimate and Score Practice) before Lesson 4. Keep this page: the lab uses your estimate and actual mass held. Engineer's Logbook: do the 15-Minute Home Kit Build (p. 15) and check it fits the 15 x 15 x 2.5 cm box for the 15-Minute Builder badge.

Mock Tournament

TODAY I CAN

- Everything on my can-do checklist (the page right after the divider). Today is the time to show it.

Mock tournament. Build first, then take this test until you are called to test your barge.

Today's steps: run it yourselves (27 minutes)

You need: 2 full practice kits (face down), measuring box, about 200 hex nuts, 0.01 g scale, 4 pairs of Category C goggles (use your own if you have them), strainer and towels, Lesson 4 answer card (Event Captain keeps it face down). Set the timer for each step.

1. Event Judge 2 min. Set the timer for 2 minutes. Read the rules out loud: 15 minutes, kit materials only, one open cargo area, must fit in the 15 x 15 x 2.5 cm box. Hand out kits face down.
2. Both 14 min. Set the timer for 14 minutes. Build with no help. The Event Judge calls 10, 5, and 1 minute left.
3. Both 1 min. Write your estimate (to 0.01 g) in 'Score box' before any testing.
4. Event Judge 10 min. Set the timer for 10 minutes. Pairs start the 'Practice test'. Call one pair at a time to test: box check, goggles on, then get an adult in the room to say 'Clear!' before testing. Load nuts one at a time, strain, weigh, and record. The pair goes back to the test.

Check your work: Ask your Event Captain for the Lesson 4 answer card. Mark your own test, fill 'Test score' and 'Final score' in the 'Score box', and redo any question you missed.

Practice test

1. What is the name for the upward push of water on an object?

2. When you put a rock in a cup of water, the water level rises. What is this called?

3. State Archimedes' principle in your own words.

4. A floating toy boat pushes aside 150 mL of water. What is the mass of the boat?

5. A barge hull is 10 cm x 10 cm x 2.5 cm. What is its volume? How many grams of water can it push aside?

6. A solid steel nut sinks. Is steel denser or less dense than water?

7. Why does a huge steel ship float?

8. When is a barge counted as sunk at the tournament?

9. What happens to how low a boat sits in the water as you add cargo?

10. A barge will not fit in the 15 x 15 x 2.5 cm box. What tier is it in?

Score box

Item	Our result
Estimate (to 0.01 g)	
Fits box, open top, kit only? (Tier 1 or 2)	
Actual mass held (g)	
Difference from estimate (g)	
Test score (out of 10)	
Bonus (5 if exact)	
Final score	

Tournament day checklist

- [ ] Category C indirect-vent goggles
- [ ] Scissors
- [ ] Ruler
- [ ] Pencils
- [ ] Calculator (optional)
- [ ] Our build steps memorized
- [ ] Mass of 1 nut and our real-world percent memorized

At-home practice

Answer these at home. Check them with the answer card at the next Huddle.

1. A pair gets 14 test points. Its barge holds 312.25 g, and its estimate was 300.00 g. What is its final score?

2. When is a barge counted as sunk?

3. A pair adds a paper clip from home to make its barge stronger. What tier is the barge in?

4. A plastic toy has a mass of 40 g and a volume of 50 cubic cm. Will it float or sink? Why?

5. Water leaks into your barge during the test. May you pour it out?

At home before next time: Answer the At-home practice questions on this handout. Redo any test question you missed and check it with your flashcards and your Independent Lab I1 and I2 pages. Practice your build steps with paper one more time. Put your goggles, scissors, ruler, and pencils in a bag for tournament day. Engineer's Logbook: do Estimate to the 0.01 g (p. 16) and score yourself like the judges for the Estimate Expert badge, then use the Tournament-day checklist (p. 17).

Lab I1: The Push of the Water

THIS LAB HELPS ME

- I can define buoyancy (buoyant force) and describe the upward push of water.
- I can define displacement and measure it in mL with a measuring cup.
- I can explain flotation: an object floats when the water it pushes aside weighs as much as it does before it goes under.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Water pushes up on everything you put in it. In this lab you feel that push and measure it. Do it alone or with your partner, at home or at the Open Lab table. Wipe up spills as you go.

Steps (27 minutes)

You need: bowl or tub of water, empty bottle with cap, measuring cup with mL marks, clay (about 30 g), 5 small waterproof things, small plastic container and 10 coins, towel. Set the timer for each step.

1. Me 4 min. Set the timer for 4 minutes. Sink or Float: for each of your 5 things, write a guess, then test it in the water. Fill 'Sink or Float'.
2. Me 4 min. Set the timer for 4 minutes. Read 'Words to know'. Push the capped empty bottle straight down into the water and feel it push back: that is buoyant force. Does it push harder when more of the bottle is under?
3. Me 8 min. Set the timer for 8 minutes. Fill the measuring cup to 300 mL (or 200 mL if it is small). Drop in a tight clay ball and read the level at eye level. Take it out, refill to the start line, shape the clay into a boat, float it, and read again. Fill 'Displacement lab'.
4. Me 4 min. Set the timer for 4 minutes. Float the plastic container in the bowl. Add coins one at a time and watch how low it sits. Write the number of coins when water almost comes over the edge.
5. Me 4 min. Set the timer for 4 minutes. Read 'Why it works' and answer 'Think about it'.
6. Me 3 min. Set the timer for 3 minutes. Check with the I1 answer card. Dry everything and put it back.

Check your work: Get the I1 answer card from the Open Lab bin (or ask your Event Captain at Lesson 2). Your numbers can be a little different. Check 'Think about it' and fix any you missed in a different color.

Sink or Float

Object	My guess (float/sink)	What happened

Words to know

Buoyancy (buoyant force)	The upward push of water (or any liquid) on an object in it.
Displacement	Water being pushed aside by an object. The water level rises.
Float	Stay on top. The upward push is as big as the object's weight.
Sink	Go down. The object's weight is bigger than the upward push.
mL (milliliter)	A unit for measuring water. 1 mL of water has a mass of 1 gram.

Displacement lab

Object	Start level (mL)	New level (mL)	Rise (mL)	Float or sink?
Clay ball				
Clay boat				

Why it works

A floating object sinks in until the water it pushes aside weighs as much as it does. A 30 g clay boat pushes aside about 30 mL of water (30 g). A tight clay ball can only push aside water the size of the ball. That water weighs less than the clay, so the ball sinks.

Think about it

1. Which pushed aside more water, the clay ball or the clay boat?

2. What happened to the container as you added coins? Why?

3. How does a boat help heavy things stay on top of the water?

Bring to the next meeting: Bring your Sink or Float and Displacement lab tables to Lesson 2.

Lab I2: Archimedes, Density, and Folding

THIS LAB HELPS ME

- I can define buoyancy (buoyant force) and describe the upward push of water.
- I can state Archimedes' principle: the upward push equals the weight of the water pushed aside.
- I can explain flotation: an object floats when the water it pushes aside weighs as much as it does before it goes under.
- I can explain why a steel ship floats but a steel nut sinks, using shape and density.
- I can use 1 cubic cm of water = 1 mL = 1 gram to predict how much cargo a barge can hold.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Archimedes found the rule that tells how much your barge can hold. In this lab you use it on your own barge, find out why steel ships float, and practice folding a fast hull.

Steps (28 minutes)

You need: 1 cm cube, bowl or tub of water, wood piece, hex nut, clay, 2 paper sheets and a ruler, your Lesson 2 handout, calculator (optional). Set the timer for each step.

1. Me 4 min. Set the timer for 4 minutes. Read 'The big rules' and 'Flotation facts' out loud. Hold the 1 cm cube: that much water is 1 mL and weighs 1 gram.
2. Me 5 min. Set the timer for 5 minutes. Answer 'Volume practice'.
3. Me 5 min. Set the timer for 5 minutes. Fill 'Our barge's capacity' using the measurements and cargo from your Lesson 2 page. Real-world percent = actual grams held / predicted grams x 100.
4. Me 5 min. Set the timer for 5 minutes. Put the wood, the nut, a clay ball, and a clay boat in the water. Fill 'Density test'.
5. Me 6 min. Set the timer for 6 minutes. Folding practice: fold a paper tray about 13 x 13 cm with 2 cm walls. Fold the corners (tuck the extra into a triangle); do not cut. Time it. Do it twice and write both times.
6. Me 3 min. Set the timer for 3 minutes. Sketch Design A or B in the box. Then check with the I2 answer card.

Check your work: Get the I2 answer card from the Open Lab bin (or ask your Event Captain at Lesson 3). Check 'Volume practice' and 'Density test', and compare your capacity steps to the example. Fix any you missed in a different color.

The big rules

Archimedes' principle	The upward push on an object equals the weight of the water it pushes aside.
The gram rule	1 cubic cm of water = 1 mL = 1 gram.
Volume	Length x width x height. Measure in cm to get cubic cm.
Capacity	Volume of the hull below the rim (in g) minus the mass of the barge = most cargo it can hold.
Real-world percent	Actual grams held / predicted grams x 100. Real barges leak, tip, and bend, so they hold less than the math.

Flotation facts

Flotation	The ability to float. An object floats when the buoyant force equals its weight before it is all the way under.
Density	How much mass is packed into a space. Heavy for its size = dense.
Denser than water	Sinks (steel, rocks, clay balls).
Less dense than water	Floats (wood, oil, a boat full of air).
Why ships float	A ship's hull holds a lot of air, so the ship plus the air is less dense than water.

Volume practice

1. A tray is 10 cm x 10 cm x 2 cm. What is its volume? How many grams of water can it push aside?

2. A tray is 12 cm x 12 cm x 2 cm and weighs 10 g. What is the most cargo it can hold?

3. The biggest barge allowed is 15 x 15 x 2.5 cm. What is its volume?

4. A barge was predicted to hold 278 g but held 222 g. What is its real-world percent?

Our barge's capacity (from Lesson 2)

Step	Our barge
Inside L x W x wall height (cubic cm)	
Most water it can push aside (g)	
Minus barge mass (g) (use 10 g if you did not weigh it)	
Predicted cargo (g)	
Actual cargo held (g)	
Real-world percent (%)	

Density test

Object	Float or sink?	Denser or less dense than water?
Wood piece		
Hex nut or metal spoon		
Clay ball		
Clay boat (clay plus air)		

Folding practice

Try	Time (min:sec)	What slowed me down
1		
2		

Sketch Design A or B for Lesson 3 (label where the tape goes)

A: plastic sheet tray lined or wrapped in foil. B: foil hull with straws taped on the rim.

Bring to the next meeting: Bring this page, with your real-world percent, and your folding times to Lesson 3.

Lab I3: Estimate and Score Practice

THIS LAB HELPS ME

- I can use 1 cubic cm of water = 1 mL = 1 gram to predict how much cargo a barge can hold.
- I can make a cargo estimate to 0.01 g using my barge's size, my practice results, and the mass of one nut.
- I can explain the score: test points plus grams held plus a 5-point bonus for an exact estimate.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. At the tournament you write down how much your barge will hold, to 0.01 g. An exact estimate earns 5 bonus points, and a close one can win a tie. This lab is a practice set you can do alone.

Steps (28 minutes)

You need: calculator, 6 index cards, your Lesson 3 handout. Set the timer for each step.

1. Me 4 min. Set the timer for 4 minutes. Read 'How Ship Shape is scored' out loud.
2. Me 8 min. Set the timer for 8 minutes. Fill the blanks in 'Estimate practice', one row at a time. Use a calculator if you want.
3. Me 5 min. Set the timer for 5 minutes. Answer 'Score and tier'.
4. Me 5 min. Set the timer for 5 minutes. Write your team's best barge as numbered steps in 'Our build steps'. Cross out any step that wasted time in Lesson 3.
5. Me 6 min. Set the timer for 6 minutes. Make 6 flashcards: buoyancy, displacement, Archimedes' principle, flotation, density, and the gram rule. Quiz yourself twice. Then check this page with the I3 answer card.

Check your work: Get the I3 answer card from the Open Lab bin (or ask your Event Captain at Lesson 4). Check every row of 'Estimate practice' and each 'Score and tier' answer. Fix any you missed in a different color.

How Ship Shape is scored

Final score	Test points + grams of cargo held + 5 bonus if the estimate is exact.
Tier 1	Barge follows all rules: open top, fits in the box, kit materials only.
Tier 2	Broke a building rule. Still tests, but ranks below every Tier 1 team.
Sunk	The top edge of the cargo area is all the way under water.
Top limit	Loading stops at 500 g.

Estimate practice

Step	Numbers
Mass of 10 nuts	32.10 g
Mass of 1 nut (divide by 10)	
Inside size	13 cm x 13 cm x 2 cm
Volume (cubic cm)	
Barge mass	12 g
Volume minus barge mass (g)	
Real-world percent	80 percent
Predicted cargo (g) (x 0.8)	
Predicted nuts (divide by mass of 1 nut, round down)	
Estimate = nuts x mass of 1 nut (to 0.01 g)	

Score and tier

1. A pair gets 18 test points, its barge holds 260.01 g, and its estimate was 250.00 g. What is its final score?

_____

2. A pair estimates 150.00 g, holds exactly 150.00 g, and gets 15 test points. What is its final score?

3. A barge holds 300 g but sticks out of the box. Why might a barge that held 200 g beat it?

4. When is a barge counted as sunk?

Our build steps (numbered)

Bring to the next meeting: Bring your flashcards and your numbered build steps to Lesson 4.